



Experimental study of the complexation of silicon and germanium with aqueous organic species: Implications for germanium and silicon transport and Ge/Si ratio in natural waters

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(Received March 9, 1998; accepted in revised form July 24, 1998)

Abstract—The stability of aqueous complexes formed by Si and Ge with carboxylic acids (acetic, salicylic, oxalic, citric, tartaric) and phenols (phenol and catechol) has been investigated from 25 to 90°C via solubility and potentiometric measurements. Results show that Ge forms stable complexes with the di- and tricarboxylic acids and catechol, but that Si forms much weaker complexes with these ligands. Analysis of our results and of available literature data on Ge complexes formed with other types of aqueous organic species demonstrates that Ge forms complexes of chelate type with the following functional groups: (1) carboxylic in acid solutions ($1 \leq \text{pH} \leq 6$), (2) di-phenolic hydroxyls in neutral and basic solutions ($\text{pH} \geq 6$), and (3) alcoholic hydroxyls in very basic solutions ($\text{pH} \geq 10$). Conversely, Si forms very weak complexes with these compounds. Stability constants generated in this study for Ge- and Si-organic species have been used to approximate Ge and Si complexing with humic acids which possess the same organic functional groups as those used in this study. Our calculations show that Si-humic acid complexes are negligible in most natural waters. In contrast, the presence of humic acids can considerably affect Ge speciation in aqueous solution. For example, at $\text{pH} \geq 6$ in a solution containing and 0.1 $\mu\text{g/L}$ of Ge and 20 mg/L of dissolved organic carbon (DOC), Ge-humic acid complexes account for more than 95% of total aqueous Ge. These results can explain the increase of the Ge/Si ratio in organic-rich surficial waters. Ge-organic matter complexation should be thus taken into account when using Ge/Si ratios measured in surface waters and biogenic opals to estimate chemical-weathering intensity and Ge and Si global fluxes. Copyright © 1998 Elsevier Science Ltd

1. INTRODUCTION

Soluble organic matter exerts a major influence on water-rock interactions (i.e., see Freise, 1931; Graham, 1941; Thurman, 1985; Hennet et al., 1988; Fein, 1994; etc.). The formation of aqueous and surface metal-organic complexes enhances both the solubility of minerals and their dissolution kinetics, thus increasing the intensity of weathering and diagenesis. Although many studies have been devoted to quantifying complexes formed by metals with organic ligands, little is known about the organic complexation of metalloids, such as Si, Ge, As, B, and Sb. The purpose of this paper is to determine the stability of Si and Ge complexes with aqueous organic ligands and to evaluate the effect of this complexation on Si and Ge transport and the Ge/Si ratio in superficial waters.

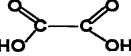
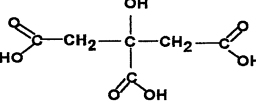
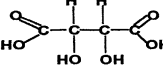
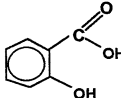
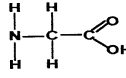
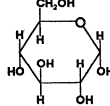
Silicon is one of the most abundant constituents of terrestrial rocks and fluids and an important biogenic element. Available data on its complexation with natural organic ligands, however, are still scarce and contradictory. For example, based on UV and Raman spectroscopic observations, and an increase of quartz dissolution rates in the presence of carboxylic acids, Bennett and Siegel (1987), Bennett et al. (1988), Marley et al. (1989), Bennett (1991), and Hiebert and Bennett (1992) suggested strong chelate complex formation between aqueous silica and oxalic and citric acids. Conversely, potentiometric and solubility measurements (Clark and Waddams, 1957; Öhman et al., 1991; Fein and Hestrin, 1994; Poulson et al., 1997) suggest

only very weak complexation between silica and carboxylic acids or alcohols. On the other hand, solubility and potentiometric measurements indicate silica forms stable complexes with di-ortho-hydroxy aromatic compounds such as tropolone, catechol, and its derivatives (Iler, 1979; Sjöberg et al., 1985; Öhman et al., 1991; Sedeh et al., 1992, 1993). As the binding sites of aqueous humic acids contain similar di-ortho-phenolic functional groups, they can complex silica and, therefore, increase its mobility in organic-rich waters. Clearly, much controversy persists around the degree of organic-aqueous silica interactions in natural waters.

Germanium has been considered as a chemical analog of Si following a similar inorganic geochemical pathway (Froelich et al., 1985a,b; Froelich and Andreae, 1981; Mortlock and Froelich, 1987; Lewis et al., 1988; Murnane and Stallard, 1990). However, there is very limited knowledge on the influence of dissolved organic matter on Ge behavior in surficial waters. Many studies have been reported in chemical literature on aqueous Ge complexes with polyalcohols (e.g. Antikainen, 1959; Antikainen and Rossi, 1959; Nazarenko and Flyantikova, 1963a,b), carboxylic acids (Moeller and Nielsen, 1953; Mattock, 1954; Vartapetian, 1957; Clark and Waddams, 1957), phenol, and catechol derivatives (Antikainen and Malkonen, 1959; Andrianov and Nazarenko, 1963; Beauchamp and Benoit, 1967; Choux and Benoit, 1967; Tsau et al., 1967). Most of them, however, were conducted in solutions containing high concentrations of both Ge and organic ligands (>0.05 – 0.1 M) in which the chemical speciation is poorly known. Moreover, the stability constants generated at 25°C for the same complex often differ by 1–2 orders of magnitude (Sillen and Martell,

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Table 1. Structures of the organic compounds used in this study and stepwise dissociation constants ($-\log K$) of their functional groups (according to Sillen and Martell, 1964, 1971)

Acetic acid (HAc)		$pK(\text{CO}_2\text{H}) = 4.76$
Oxalic acid (H ₂ Ox)		$pK_1(\text{CO}_2\text{H}) = 1.25$ $pK_2(\text{CO}_2\text{H}) = 4.27$
Citric acid (H ₄ Cit)		$pK_1(\text{CO}_2\text{H}) = 3.13$ $pK_2(\text{CO}_2\text{H}) = 4.76$ $pK_3(\text{CO}_2\text{H}) = 6.40$ $pK_4(\text{OH}) \sim 16$
Tartaric acid (H ₄ Tart)		$pK_1(\text{CO}_2\text{H}) = 3.04$ $pK_2(\text{CO}_2\text{H}) = 4.37$ $pK_3(\text{OH}) \sim 14$ $pK_4(\text{OH}) \sim 15.5$
Salicylic acid (H ₂ Sal)		$pK(\text{CO}_2\text{H}) = 2.97$ $pK(\text{OH}) = 13.60$
Glycine (HGly)		$pK(\text{CO}_2\text{H}) = 9.78$
Glucose (Glu)		
Phenol (HPhen)		$pK(\text{OH}) = 9.80$
Catechol (H ₂ Cat)		$pK_1(\text{OH}) = 9.45$ $pK_2(\text{OH}) = 13.0$

1971). There is also very little knowledge on the effect of temperature on these values.

In this paper we present solubility and potentiometric measurements at 25 and 90°C to quantify the interactions of Si and Ge with aqueous organic species. This study was performed with simple organic ligands containing the same functional groups as those encountered in natural organic matter: aliphatic carboxylic acids (acetic and oxalic), aromatic carboxylic acid (salicylic), hydroxy-carboxylic acids (citric and tartaric), amino acid (glycine), polyalcohol (glucose), phenol, and di-phenol (catechol) (Table 1). In addition, several solubility measurements were performed in solutions containing a humic acid extracted from a soil. The results presented in this study allow quantification of the effect of dissolved organic matter on Ge and Si behavior in surficial aqueous environments.

2. MATERIALS AND METHODS

2.1. Solid Materials and Solution Preparation

Solubility measurements of amorphous silica were conducted with a hydrated silica gel powder (Baker Analyzed Reagent) from which small particles were removed by sedimentation in deionized water. About 50 g of this powder were stirred in a 2 dm³ glass beaker filled with de-ionized water until the entire solid was suspended. The bulk of solid sedimented within 10 min after which the supernatant containing fine particles was withdrawn. The washing was repeated until the supernatant became transparent within 1–2 min. The powder was then

oven dried at 60°C for 24 h. X-ray powder analysis of this material showed no crystalline phases. The powder contained ~40% wt H₂O as shown by thermogravimetric analysis. Its specific surface area was $450 \pm 25 \text{ m}^2\text{g}^{-1}$ as measured by the BET method using Nitrogen as the adsorbing gas.

Germanium oxide solubility measurements were performed with a powder of commercially available GeO₂(hex) (99.999%, Aldrich). X-ray diffraction analysis showed that this material was pure well-crystallized hexagonal Ge dioxide (JCPDS card 4-0497). Its specific surface area was $0.5 \pm 0.05 \text{ m}^2\text{g}^{-1}$ as measured by the BET method using Krypton as the adsorbing gas.

Experimental solutions used for solubility measurements were prepared by weight from deionized water and the following reagent grade organic compounds: acetic acid, oxalic acid, citric acid, tartaric acid, salicylic acid, glycine, glucose, phenol, and catechol. The humic acid powder (20% of ash, Fluka) was dissolved in 0.1 M NaOH solution. The insoluble fraction was separated by ultra centrifugation (10 min at a rotation speed of 5000 revs/min), and filtration through a 0.45 μm Millipore membrane. The concentration of dissolved organic carbon (DOC) in the filtered solution was 1g/L, as determined by a DORHMAN DC-80 carbon analyzer. The experimental solutions contained up to 0.1 m of each organic compound and 1g/L of COD in the case of humic acid. Ionic strengths and pH obtained by adding NaCl, NaOH (1 M), or HCl (1 M) covered the ranges 0.1–0.3 m and 2–11, respectively. Blank solutions without organic compounds were prepared with NaCl, B(OH)₃, NaOH, or HCl and had the same Na concentrations (0.1–0.3 m) and pH as the organic-bearing solutions.

Experimental solutions of citric and oxalic acids and catechol used for potentiometric measurements were prepared from the corresponding organic reagents using NaCl, HCl, NaOH, or KOH to adjust pH and ionic strength. Silicic acid (Si(OH)₄) solutions were obtained by dissolution of the silica gel in deionized water at 250°C. Before each experiment, the solution was sampled and filtered through a 0.45 μm Millipore membrane. Germanic acid solutions (Ge(OH)₄) were prepared by dissolution of GeO₂(hex) in deionized water at ambient temperature, and then were filtered through a 0.45 μm Millipore filter.

2.2. Analytical Methods

Aqueous Ge concentrations were measured by flame atomic absorption spectroscopy (AAS) or by colorimetry. Germanium standard solutions for both methods were prepared by weight dilution of an Aldrich 1005 ppm Ge standard solution. For the concentration range 10–100 ppm, Ge was analysed by AAS at an absorption line of 261.5 nm using acetylene-nitrogen protoxide flame (Pokrovski, 1996). Reproducibility of this method was better than 5%, and the detection limit defined at 2σ standard deviation was found to be 0.5 ppm. Analyses of standard solutions in the presence of the organic reagents used in the experiments showed no influence on the calibration curve. Lower Ge concentrations, between 1 and 20 ppm, were analysed by the molybdate blue method using ascorbic acid as reducing agent. The analyses were performed on an Autoanalyser-II Technicon following the protocol for Si analysis (see below). No influence of the organic matrix was detected. The reproducibility and detection limit of Ge analyses were 5% and 0.1 ppm, respectively. Germanium concentrations of the same experimental solutions analyzed by AAS and colorimetry were the same in the limit of 5%.

Aqueous Si concentrations were measured by colorimetry or by ICP-MS. Monomeric silica was analyzed in the concentration range 0.1–10 ppm by the molybdate blue method (Koroleff, 1976) using an automatic chain Autoanalyser-II Technicon. Reproducibility and detection limit were 2% and 0.05 ppm, respectively. No significant matrix effects were found for any of the organic solutions. The validity of colorimetric analyses was checked by ICP-MS determination of total dissolved silica, using the ²⁹Si isotope. The reproducibility and detection limit for the ICP-MS analysis were 5% and 0.1 ppm of Si, respectively. Concentrations in the same solutions measured by ICP-MS and colorimetry were always the same in the limit of ±3%.

2.3. Solubility Measurements

Solubility measurements were conducted at $25 \pm 1^\circ\text{C}$ in 60–150 cm³ polypropylene bottles immersed in a water thermostated bath and

gently shaken on a shaker. Starting solution/solid ratio ranged from 4 to 1100 and from 30 to 100 for silica and Ge, respectively. To monitor attainment of equilibrium, 5 to 10 mL of solution were taken periodically, filtered through a 0.45 μm Millipore membrane, diluted with deionized water, and analyzed for Si or Ge as described above. pH of experimental solutions was measured in situ during each sampling, using a combination glass electrode (SCHOTT H61) calibrated on the activity scale using the NBS/DIN phthalate ($\text{pH}_{25^\circ\text{C}} = 4.008$), phosphate ($\text{pH}_{25^\circ\text{C}} = 6.865$) and borate ($\text{pH}_{25^\circ\text{C}} = 9.180$) buffer solutions.

2.4. Potentiometric Measurements

Potentiometric measurements were performed at 25 and 90°C on aqueous citrate and oxalate-bearing solutions in the presence of silicic acid, and on citrate and catechol-bearing solutions in the presence of germanic acid. pH was measured using a combination glass electrode (SCHOTT H61) connected to a Philips PW9422 digital pH-meter with a 0.1 mV resolution. The electrode was fixed in the plastic cover of a $\sim 60\text{ cm}^3$ Pyrex, water thermostated ($\pm 0.2^\circ\text{C}$) potentiometric cell equipped with a magnetic Teflon-coated stirring bar. The electrode calibration was performed using 0.1 M HCl ($\text{pH}_{25^\circ\text{C}} = 1.10$), 0.01 M HCl ($\text{pH}_{25^\circ\text{C}} = 2.04$) solutions, and the NBS/DIN phthalate ($\text{pH}_{25^\circ\text{C}} = 4.008$), phosphate ($\text{pH}_{25^\circ\text{C}} = 6.865$), and borate ($\text{pH}_{25^\circ\text{C}} = 9.180$) buffer solutions. Good linear relationships between the electrode potential and solution pH were obtained to 90°C (correlation coefficient > 0.9999) with the electrode slopes corresponding to the theoretical Nernst slopes in the limit of 2%.

Two types of titration procedures were used. The first is a KOH or NaOH titration of organic-bearing solutions with and without silica or Ge. These measurements permit to cover a wide range of H^+ concentrations, enabling determination of the pH intervals in which complexation occurs. The second type of measurement consisted of titrations by Ge- or Si-bearing aqueous solution of organic-compound solutions at the desired pH and temperature. This technique is more sensitive than the KOH/NaOH titration and enables detection of very small pH changes (~ 0.002 pH units) due to complexation. The e.m.f. of the potentiometric cell stabilized in 0.5–1 min after each injection of the titrant solution and remained constant (± 0.2 mV) for at least 0.5 h. During the experiments, solution aliquots were periodically taken from the cell using a plastic syringe, filtered through a 0.45 mm membrane, diluted and analyzed for dissolved silica or Ge. Measured Ge and Si concentrations were always within $\pm 2\%$ of those calculated from the initial and injected fluid compositions. This demonstrates that neither silica or germanic acid polymerization, nor precipitation of Si/Ge-organic-bearing solids occurred during the experiments.

3. RESULTS

3.1. Complexation of Silica with Aqueous Organic Species

3.1.1. Potentiometric measurements

Potentiometric measurements conducted at 25 and 90°C at pH ranging from 2 to 10 in solutions containing up to 0.006 M of silicic acid and 0.005 M of citric or oxalic acid did not show any changes in pH after Si injection, nor modifications of KOH-titration curves of citric and oxalic acids in the presence of Si. These results are in agreement with those of Öhman et al. (1991) and Poulson et al. (1997) who obtained identical curves during NaOH-titrations of oxalate-bearing solutions (0.005–0.4 M) with and without aqueous silica (0–0.4 M).

3.1.2. Solubility measurements

The results of amorphous silica solubility experiments at 25°C are reported in Table 2 as a function of solution composition, pH and dissolution time. It can be seen in this Table that with solution/solid ratios (R) from 30 to 100 most of the runs performed at $\text{pH} \geq 6$ –7 in NaCl+HCl/NaOH/NaBO₂ solutions

without organic compounds approach to steady-state in less than 10 days. In neutral and weakly acid media the approach of steady-state is slower (20–30 days). In acid solutions ($\text{pH} \leq 3$) with R from 30 to 100, steady-state was not attained even after 2 months (run A). This influence of pH on silica dissolution rate is at least qualitatively consistent with the results of Wirth and Gieskes (1979), Guy and Schott (1989) and Cadoré (1995). At much lower solution/solid ratios ($R = 4$), steady-state was reached in a week, even at acid pH (runs B, C). For runs conducted at neutral and basic pH, a slight decrease in Si concentration has been observed after one month. This is likely to reflect an ageing process that causes silica particles to cement together, thus decreasing their solubility (Rimstidt and Cole, 1983; Mizele et al., 1985). Silicon concentration measured at steady-state in solutions of neutral pH is $51 \pm 2\text{ mg/L}$ (runs D, E, F, I). This value is in agreement with that proposed by Walther and Helgeson (1977) for the solubility of amorphous silica (54 mg/L at $I = 0$) and slightly lower than that reported by Marshall and Warakowski (1980) in the presence of 0.1 M NaCl (60 mg/L). It can be seen in Table 2 that the increase of the solution/solid ratio from 5 to 1000 diminishes silica dissolution rate but does not affect Si concentration at steady-state. As a result, the attainment of dissolution steady-state was used as the equilibrium criterion.

For the experiments conducted in NaCl+HCl/NaOH solutions containing up to 0.1 M of carboxylic acids, glycine, glucose, or phenol (runs J–AF), Si concentrations measured at the same pH after the same dissolution time or at steady-state are identical to those measured in the corresponding organic-free runs. The logarithm of aqueous Si molality in equilibrium with amorphous silica in solutions containing 0.1 M of one of these organic ligands is plotted as a function of pH in Fig. 1. It can be seen that the experimental points fall on the dashed curve which depicts calculated silica solubilities in NaCl-HCl-NaOH solutions using amorphous silica solubility product determined in this study at neutral pH and silicic acid ionization constants reported in Table 3. The measurements reported in this study thus indicate that Si complexes with carboxylic acids, amino acid, polyalcohol, and phenol are not significant in the limit of the experimental uncertainties ($\pm 2\text{ mg/L}$ of Si). These results are close to those of Öhman et al. (1991) who reported a $\sim 5\text{ mg/L}$ increase of Si concentration in a 0.6 M NaCl–0.1 M citrate or oxalate solution equilibrated at pH 7 with amorphous silica.

By contrast, in the presence of 0.1 M of catechol at pH between 8 and 10, silica solubility increases by an order of magnitude (Table 2, runs AH–AL; Fig. 1). This increase is accompanied by a significant solution pH change. In most runs solubilities attain a maximum after 1–2 weeks and then decrease slightly with time. This decrease is also accompanied by a pH decrease which is likely to be caused by the oxidation of catechol. However, calculations using these pH changes suggest that the concentration of the oxidation products formed are negligible compared to the total catechol concentration ($< 5\%$). The increase of silica concentration and change of solution pH in comparison with catechol-free solutions is likely caused by the formation of a hexacoordinated Si-catechol complex (Iler, 1979; Öhman et al., 1991) according to the reaction:

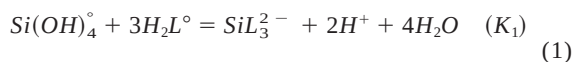


Table 2. Aqueous Si concentrations measured as a function of pH and time during amorphous silica dissolution in NaCl + HCl (or NaOH) solutions (total Na ranging from 0.1 to 0.3 m) in the presence or not of 0.1 m of organic compound

Solution composition and initial solution to solid ratio	Run	Time (days)	pH	C _{Si} (mg/L)
0.1 m NaCl + HCl	100:1	A	3	2.99
			10	2.99
			66	3.05
0.1 m NaCl + HCl	4:1	B	8	2.64
			35	2.60
0.1 m NaCl + HCl	4:1	C	8	3.04
			35	3.05
0.1 m NaCl	96:1	D	3	5.79
			11	5.74
			18	5.74
			33	5.75
0.1 m NaCl + NaOH	32:1	E	2	6.61
			8	6.16
			38	6.16
0.1 m NaCl + NaOH	97:1	F	3	8.44
			11	8.32
			18	8.29
			33	8.25
0.1 m NaCl + NaOH	95:1	G	3	9.87
			10	9.89
			22	9.87
0.1 m NaCl + NaBO ₂	100:1	H	3	9.16
			10	9.18
			22	9.18
0.3 m NaCl + NaOH	658:1	I	2	6.53
			9	6.03
			21	6.33
			35	6.32
			133	6.24
0.1 m glycine + 0.1 m NaCl + HCl	100:1	J	3	2.96
			11	2.97
			18	2.95
			33	2.96
0.1 m glycine + 0.1 m NaCl	100:1	K	3	5.84
			11	6.05
			18	6.24
			33	6.54
0.1 m glycine + 0.1 m NaCl + NaOH	95:1	L	3	9.58
			11	9.52
			18	9.52
			33	9.56
0.1 m salicylic acid + NaOH	95:1	M	3	4.82
			11	4.81
			18	4.81
			33	4.84
0.1 m salicylic acid + NaOH	95:1	N	3	8.11
			11	8.02
			18	8.02
0.1 m salicylic acid + NaOH	90:1	O	3	9.98
			10	9.96
			22	9.96
0.1 m salicylic acid + NaOH	1100:1	P	2	9.78
			9	9.73
			21	9.68
			21	9.68
0.1 m oxalic acid + NaOH	5:1	Q	5	2.83
			8	2.81
			35	2.80
0.1 m tartaric acid + NaOH	5:1	R	5	3.50
			8	3.50
			35	3.50
0.1 m tartaric acid + NaOH	50:1	S	7	3.55
			14	3.51
			21	3.50
			31	3.50

continued

Table 2. (Continued)

Solution composition and initial solution to solid ratio		Run	Time (days)	pH	C _{Si} (mg/L)
0.1 m tartaric acid + NaOH	80:1	T	3	5.94	33.24
			10	6.04	49.59
			22	6.16	51.15
			36	6.32	50.89
0.1 m tartaric acid + NaOH	95:1	U	3	9.39	74.13
			10	9.39	76.25
			22	9.39	74.38
0.1 m glucose + 0.1 m NaCl	95:1	V	3	6.27	46.8
			10	4.18	48.56
			22	3.59	49.72
0.1 m glucose + 0.1 m NaCl + NaOH	104:1	W	4	9.42	74.35
			14	9.41	72.21
0.1 m citric acid + NaOH	6:1	X	5	3.69	49.01
			8	3.69	47.57
			35	3.69	47.09
0.1 m citric acid + NaOH	50:1	Y	7	3.83	33.14
			14	3.76	39.20
			21	3.72	41.41
			31	3.68	42.39
0.1 m citric acid + NaOH	96:1	Z	3	6.90	47.74
			10	6.91	51.78
			22	6.92	52.55
			36	6.92	51.08
0.1 m citric acid + NaOH	90:1	AA	133	7.03	49.40
			3	9.64	101.0
			10	9.66	101.1
0.1 m citric acid + NaOH	650:1	AB	19	9.63	95.43
			2	9.63	66.41
			9	9.58	93.33
0.1 m citric acid + NaOH	960:1	AC	21	9.54	90.56
			2	6.95	34.68
			9	6.97	52.23
			21	6.96	52.42
0.1 m phenol + NaOH	53:1	AD	35	6.96	50.58
			3	10.52	288
			10	10.48	346
0.1 m phenol + NaOH	102:1	AE	17	10.48	331
			4	9.44	75.41
			11	9.44	78.89
0.1 m phenol + NaOH	51:1	AF	3	7.41	51.16
			10	7.43	49.31
			17	7.43	48.39
0.1 m catechol + NaOH	103:1	AG	4	4.03	31.34
			11	3.73	32.93
			18	3.66	32.47
0.1 m catechol + NaOH	95:1	AH	4	7.74	114.9
			11	7.18	146.3
			18	7.05	111.0
0.1 m catechol + NaOH	52:1	AI	3	8.82	343
			10	8.09	490
			17	7.95	407
			24	7.81	316
0.044 m catechol + NaOH	53:1	AJ	3	9.58	283
			10	9.20	327
			17	8.85	264
			24	8.55	212
0.1 m catechol + NaOH	100:1	AK	4	9.44	460
			11	8.94	757
			18	8.78	772
0.1 m catechol + NaOH	55:1	AL	3	10.60	502
			10	10.31	958
			17	10.03	821
			24	9.81	819

continued

Table 2. (Continued)

Solution composition and initial solution to solid ratio	Run	Time (days)	pH	C _{Si} (mg/L)
Humic acid 1 g/l DOC + 0.1 m NaOH + HCl	AM	2	4.82	38.25
		8	5.03	45.15
		37	5.07	50.00
Humic acid 1 g/l DOC + 0.1 m NaOH + HCl	AN	2	7.25	49.15
		8	6.94	48.90
		37	7.12	49.90
Humic acid 1 g/l DOC + 0.1 m NaOH + HCl	AO	2	7.72	49.35
		8	7.55	49.60
		37	7.46	49.90
Humic acid 1 g/l DOC + 0.1 m NaOH + HCl	AP	2	9.39	80.20
		8	9.37	73.50
		37	9.35	73.10

where H₂L stands for the catechol molecule. The stability constant for this complex (K_1) was calculated from the solubility data obtained in this study, and the dissociation constants of catechol and silicic acid reported in Table 3. The solubility and pH values used in these calculations (Table 4) represent the average values for each experiment from samples taken after 10 days of run duration (see Table 2). Activity coefficients for charged species were approximated using the extended Debye-Hückel equation with $a = 4.5 \text{ \AA}$ for all ions. Activity coefficients for neutral species were assumed to be unity. Log K_1 values derived for each experimental run are reported in Table 4. These values are independent of pH and catechol concentration in the limit of their uncertainties which strongly supports the concept that only 1:3 Si-catechol complexes formed in solution. SiL_3^{2-} complex was also characterized by ^{29}Si -NMR measurements (Sedeh, 1992). Replacement of SiL_3^{2-} by 1:2 or 1:1 Si-catechol complexes in the speciation scheme led to stability constant values which varied significantly with pH and ligand concentration. The average log K_1 value at 25°C deduced from our measurements is -12.0 ± 0.4 . This value is in agreement with that determined potentiometrically by Öhman

et al. (1991) and extrapolated to the zero ionic strength using the extended Debye-Hückel equation for activity coefficients: $\log K_1 = -11.5 \pm 0.2$. In Fig. 1 it can be seen that the solid line calculated using Öhman et al. (1991) log K_1 value is close to the experimental values obtained in this study. The formation of this complex becomes significant at $\text{pH} > 7$ (Fig. 1), therefore, suggesting that catechol-like binding sites of humic acids are potential silica complexants in natural waters. However, the high ligand concentrations required to obtain significant amounts of silica-catechol complexes ($[\text{L}] > 0.01 \text{ m}$) are unlikely to be found in natural systems. For example, in the presence of 1g/L humic acid ($[\text{L}] \sim 10^{-3} \text{ m}$; Thurman, 1985), amorphous silica solubility remains unchanged (Table 2, runs AM-AP). This is in agreement with calculations performed using K_1 value obtained in this study which show that even if all aromatic hydroxyl functional groups are catechol-like, silica solubility should increase by less than 1%.

3.2. Complexation of Germanium with Aqueous Organic Species

3.2.1. Potentiometric measurements

Catechol. pH measurements were performed at 25 and 90°C in catechol-bearing HCl-0.1 m NaCl solutions as a function of added germanic acid. Solution compositions, measured and calculated pH are reported in Table 5. The pH values of the experimental solutions were obtained from the e.m.f. values using calibration curve parameters at the corresponding temperature. The variation of liquid junction potential upon injection of aqueous germanic acid was neglected because the dominant Ge species is the neutral $\text{Ge}(\text{OH})_4^0$ and its complexation with catechol causes little change in the ionic composition of experimental solutions. The uncertainty of measured pH are estimated to be $\pm 0.005 \text{ pH units}$. The values of pH calculated by assuming that no Ge-catechol complexation occurs are also reported in Table 5. They were calculated from the solution composition using the computer code Balance (Akinfiev, 1986) which is based on minimization of the Gibbs free energy of the system. The required dissociation constants of germanic acid and catechol were taken from Table 3, and activity coefficients of charged species were calculated using the extended Debye-Hückel equation. It can be seen in Table 5 and Fig. 2 that each injection of $\text{Ge}(\text{OH})_4$ -NaCl titrant solution into the catechol-

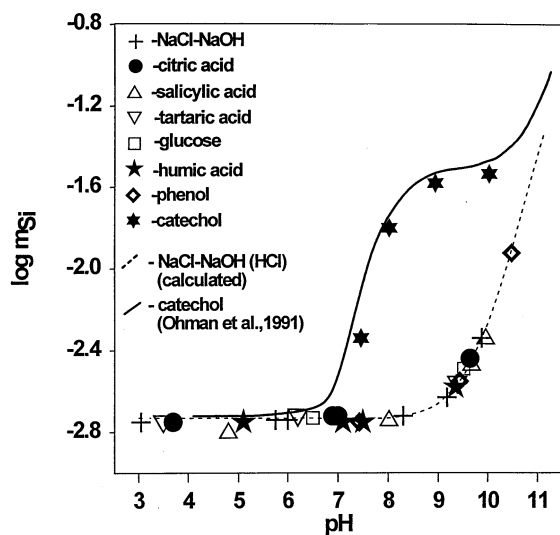
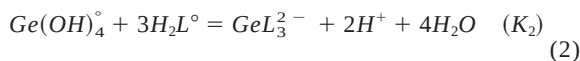


Fig. 1. Amorphous silica solubility ($\log m_{\text{Si}}$) at 25°C as a function of pH and in the presence of different organic compounds (0.1 m).

Table 3. Values at 25 and 90°C of the equilibrium constants used in this study

Reaction	log K		Reference
	25°C	90°C	
$H^+ + OH^- = H_2O$	14.00	12.42	Sweeton et al., 1974
$SiO_2(am) + 2H_2O = Si(OH)_4$	-2.74	—	This study
$SiO(OH)_3^- + H^+ = Si(OH)_4$	9.82	9.18	Busey and Mesmer, 1977
$SiO_2(OH)_2^{2-} + H^+ = SiO(OH)_3^-$	13.06	—	Baes and Mesmer, 1976
$HCat^- + H^+ = H_2Cat^0$	9.45	—	Sillen and Martell, 1971
$Cat^{2-} + H^+ = HCat^-$	13.0	—	Sillen and Martell, 1971
$HCit^{3-} + H^+ = H_2Cit^{2-}$	6.40	6.70	Bénézech et al., 1997
$H_2Cit^{2-} + H^+ = H_3Cit^-$	4.76	4.85	Bénézech et al., 1997
$H_3Cit^- + H^+ = H_4Cit^0$	3.13	3.12	Bénézech et al., 1997
$GeO_2(hex) + 2H_2O = Ge(OH)_4$	-1.38	—	This study
$GeO(OH)_3^- + H^+ = Ge(OH)_4$	9.32	8.65	Pokrovski and Schott, 1998
$GeO_2(OH)_2^{2-} + H^+ = GeO(OH)_3^-$	12.5	—	Pokrovski and Schott, 1998

bearing solution results in a significant drop of measured pH values whereas pH calculated without taking into account Ge-catechol complexes increases slightly, reflecting the dilution of the initial acid solution. Note also that the injection of a 0.1 m NaCl solution containing no Ge causes a weak increase of the measured pH which is identical to that predicted by calculations. Therefore, the difference between measured and calculated pH can be attributed to the formation of Ge-catechol complexes according to the reaction proposed by Antikainen and Malkonen (1959) and Andrianov and Nazarenko (1963):



To compute the equilibrium constant of reaction 2, GeL_3^{2-} was included into BALANCE input file, and K_2 was systematically adjusted until the calculated pH equaled the experimentally observed value. The values of $\log K_2$ calculated for each injection are reported in Table 5. These values appear to be independent, within the limit of uncertainties, of both Ge and catechol concentrations which vary by nearly an order of magnitude in different experiments. The formation of 1:1 and 1:2 Ge-catechol complexes was also tested, but neither of these stoichiometries enabled accurate reproduction of the measured pH variations. This provides further support to the speciation scheme adopted in this study. The average $\log K_2$ values at 25 and 90°C are listed in Table 5 together with their corresponding uncertainties. Our value of $\log K_2$ determined at 25°C (-1.39 ± 0.05) is in reasonable agreement with those of Antikainen

and Malkonen (1959) (-0.77 at $I = 0.1$) and Andrianov and Nazarenko (1963) (-1.79 at $I = 0$). The value of $\log K_2$ generated at 90°C ($\log K_2 = -4.05 \pm 0.1$) indicates that the stability of Ge-catechol complexes decreases rapidly with increasing temperature. The standard enthalpy ($\Delta_r H^\circ$) and entropy ($\Delta_r S^\circ$) of reaction 2 were calculated from $\log K_2$ values at 25 and 90°C assuming that $\Delta_r H^\circ$ was constant in this temperature range. The obtained values, $\Delta_r H^\circ = -20.65$ kcal/mol and $\Delta_r S^\circ = -75.65$ cal/(mol.K), are of the same sign and order of magnitude as those for the formation reactions of boron-catechol and arsenic-catechol complexes (Sillen and Martell, 1971).

Citric acid. NaOH titrations of two citric acid solutions ($m_{\text{citrate}} = 0.015$ m) with and without germanic acid ($m_{\text{Ge}} = 0.009$ m) were performed at 25°C and pH ranging from 0.5 to 10 to constrain the pH range in which complexation occurs. Results showed, that for the same added NaOH quantity, the solution containing germanic acid is more acid than its corresponding analog without Ge in the pH range 2–4 and more basic in the pH range 4–7. The difference between the titration curves for the Ge-bearing and Ge-free solutions in the pH range 2–7 implies formation of Ge-citrate complexes. To quantify this complexation more precisely, titrations of aqueous citric acid-NaOH-0.1m NaCl solutions with germanic acid were performed at 25 and 90°C following the same protocol as for the catechol-bearing solutions. Solution compositions and measured and calculated pH are reported in Table 6. The observed difference between calculated and measured pH is attributed to

Table 4. Amorphous silica solubility at 25°C as a function of pH in the presence of catechol—Values of the equilibrium constant of reaction (1)

Run	pH (25°C) average	C _{Si} (mg/l) average	m _{Si} (mol/kg)	m _{catechol} (mol/kg)	log K ₁
AH	7.46 ± 0.30	130 ± 15	0.0052	0.10	-11.8 ± 0.3
AI	8.02 ± 0.10	449 ± 50	0.016	0.10	-11.7 ± 0.2
AK	8.94 ± 0.10	747 ± 40	0.027	0.10	-12.2 ± 0.2
AJ	9.20 ± 0.30	291 ± 30	0.010	0.044	-12.1 ± 0.2
AL	10.15 ± 0.15	867 ± 50	0.031	0.10	-12.3 ± 0.2
log K₁ average					-12.0 ± 0.4
log Q ₁ (Öhman et al., 1991) in 0.6 m NaCl					-10.44 ± 0.03
log K ₁ (Öhman et al., 1991) extrapolated to zero ionic strength (this study)					-11.5 ± 0.2

Table 5. Measured and calculated pH values of solutions containing NaCl (0.1 m), HCl and catechol as a function of added germanic acid (1), and derived equilibrium constants for reaction (2)

$m_{\text{catechol}} \times 10^3$ mol/kg H ₂ O	$m_{\text{HCl}} \times 10^3$ mol/kg H ₂ O	$m_{\text{Ge}} \times 10^3$ mol/kg H ₂ O	pH measured	pH calculated without Ge-catechol complexes	log K ₂
25°C					
10.76	1.06	0	3.066	3.066	—
9.76	1.03	1.33	2.994	3.080	−1.39
9.29	0.975	3.21	2.957	3.102	−1.37
8.89*	0.933	3.07	2.979	3.121	−1.38
8.14	0.854	6.17	2.962	3.160	−1.37
7.51	0.788	8.79	2.965	3.195	−1.35
50.5	1.060	0	3.066	3.066	—
49.3	1.040	0.943	2.711	3.074	−1.45
46.6	0.978	3.09	2.509	3.101	−1.41
42.8	0.898	6.10	2.433	3.138	−1.39
39.1	0.821	9.02	2.413	3.177	−1.39
log K (25°C) average					−1.39 ± 0.05
90°C					
57.0	0.539	0	3.370	3.370	—
53.3*	0.539	0	3.371	3.370	—
51.5	0.539	1.40	3.239	3.370	−4.23
47.9	0.539	4.20	3.149	3.370	−4.16
42.0	0.539	8.70	3.097	3.370	−4.10
50.5	1.06	0	3.074	3.077	—
48.8	1.03	1.32	3.062	3.091	−4.26
45.6	0.957	3.89	3.027	3.121	−4.04
41.0	0.862	7.48	3.016	3.167	−3.95
37.7*	0.791	6.87	3.055	3.204	−3.93
log K (90°C) average					−4.05 ± 0.10

(1) Titrant solution contained 0.04 m H₄GeO₄ + 0.1 m NaCl

* Injection of a germanium-free blank solution (0.1 NaCl or 0.1 NaCl + HCl).

the formation of Ge-citrate complexes. The following rules were applied to characterize the stoichiometry and stability of these complexes: (1) The measured pH changes depend only on the complex charge and stoichiometry relative to Ge and citrate

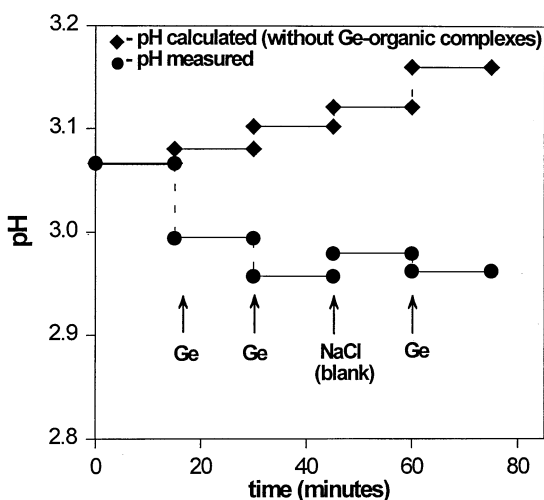


Fig. 2. Variation of pH with time during the titration of a catechol-bearing solution (0.01 m catechol + 0.1 m NaCl + 0.001 m HCl) with a Ge-bearing solution (0.04 m Ge(OH)₄ + 0.1 m NaCl). Solid circles and diamonds represent measured and calculated (without taking account of Ge-catechol complexes) pH, respectively. The arrows indicate time at which the Ge-bearing titrant solution was injected into the test solution. The difference between the measured and calculated pH shows the formation of Ge-catechol complexes.

(Ge/citrate atomic ratio). For example, Ge(OH)₃HCit^{2−} and Ge(OH)₄(H₂Cit)^{2−} which only differ by the presence of an additional water molecule in the second species, and thus display the same pH dependencies, are considered as the same complex. (2) In agreement with available data (Clark and Waddams, 1957; Evdokimov and Kogan, 1971), and by analogy with the complexes formed by Ge with tartaric, lactic and malic acids (Vartapetian, 1957), only the 1:1 and 1:2 Ge-citrate stoichiometries were considered in calculations. (3) Only complexes of a charge $Z \geq -3$ were assumed to form at acid pH (pH < 4). This is consistent with the speciation of germanic and citric acids which do not form negatively charged species in acid solutions. (4) A minimal number of Ge-citrate species is postulated to fit the measured pH changes.

The Ge-citric acid complexes obeying the above rules are listed in Table 7. The selection, among these species, of the complexes formed in our experimental solutions was performed using the code BALANCE. The stability constant value (K_i) of each complex was adjusted in order to match the measured pH values for each titration point. The variation of K_i values as a function of pH and Ge concentration was used as the selection criterion. Calculations showed that none of the complexes listed in Table 7 could alone account for the measured pH over the investigated range ($2 \leq \text{pH} \leq 7$): some complexes exhibited a variation of their stability constants by more than 3 orders of magnitude, the others led to calculated pH changes opposite to the observed ones. Then, different pairs of complexes were checked. For the majority of the tested pairs the variations in the stability constant values as a function of pH and Ge concentration were more than 2 orders of magni-

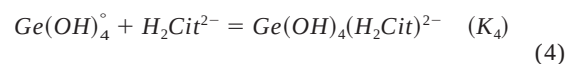
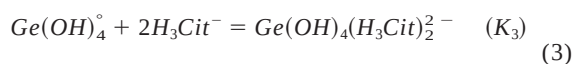
Table 6. Measured and calculated pH values of solutions containing NaCl (0.1 m), NaOH and citric acid as a function of added germanic acid (1), and derived equilibrium constants for reactions (3) and (4)

$m_{\text{citrate}} \times 10^3$ mol/kg H ₂ O	$m_{\text{NaOH}} \times 10^3$ mol/kg H ₂ O	$m_{\text{Ge(OH)}_4} \times 10^3$ mol/kg H ₂ O	pH measured	pH calculated without Ge-citrate complexes	log K ₃	log K ₄
25°C						
20.10	0	0	2.450	2.450	—	—
19.28	0	1.63	2.330	2.461	5.81	—
18.12	0	3.94	2.180	2.476	6.15	—
16.71	0	6.66	2.075	2.500	6.46	—
15.76*	0	6.28	2.097	2.510	6.46	—
14.46	0	8.95	2.077	2.530	6.46	—
19.30	39.30	0	5.050	5.047	—	—
18.47	35.70	1.67	5.161	5.058	5.63	2.60
17.60	33.90	3.54	5.313	5.048	5.78	2.32
16.30	31.50	6.11	5.563	5.062	6.37	2.32
15.40	29.60	8.12	5.691	5.046	6.54	2.06
14.60*	28.20	7.71	5.686	5.064	6.54	2.06
13.66	26.40	9.78	5.734	5.068	6.57	2.01
18.93	54.66	0	6.837	6.837	—	—
16.95*	48.92	0	6.825	6.835	—	—
15.28	44.11	3.94	7.167	6.837	—	2.55
14.02	40.50	6.91	7.215	6.844	—	2.34
13.03	37.62	9.30	7.222	6.836	—	2.22
14.45	6.80	8.87	2.39	3.04	6.15	2.49
14.27	16.40	8.78	4.21	3.93	5.85	2.79
13.98	25.60	8.69	5.57	4.85	6.46	2.14
				log K average	6.24 ± 0.30	2.36 ± 0.30
90°C						
20.40	0	0	2.470	2.470	—	—
19.90	0	1.07	2.420	2.475	4.77	—
19.00	0	2.90	2.38	2.486	4.65	—
17.70	0	5.60	2.336	2.504	4.68	—
16.40*	0	5.20	2.364	2.522	4.68	—
15.20	0	8.00	2.35	2.541	4.66	—
19.60	36.10	0	5.020	5.015	—	—
19.00	35.04	1.24	5.088	5.020	4.64	1.63
18.57	34.20	2.20	5.127	5.024	4.66	1.63
17.40*	32.10	2.06	5.134	5.030	4.68	1.59
16.32	30.07	4.57	5.215	5.037	4.72	1.55
15.32	28.20	6.90	5.264	5.042	4.61	1.50
14.70	27.09	8.30	5.278	5.050	4.61	1.50
19.00	49.15	0	6.345	6.352	—	—
18.40	47.60	1.20	6.370	6.353	—	1.68
17.40	45.10	3.30	6.400	6.363	—	1.55
16.08	41.60	6.20	6.427	6.355	—	1.59
14.90*	38.50	5.70	6.427	6.353	—	1.63
13.77	35.60	8.30	6.440	6.356	—	1.50
				log K average	4.68 ± 0.10	1.59 ± 0.10

(1) The titrant solution contained 0.04 m H₄GeO₄ + 0.1 m NaCl

* Injection of a germanium—free blank solution (0.1 m NaCl)

tude. It was found that the $\text{Ge(OH)}_4(\text{H}_3\text{Cit})_2^{2-}$ – $\text{Ge(OH)}_4(\text{H}_2\text{Cit})^{2-}$ pair provides the best fit of measured pH in the whole pH range at 25°C with the least variations of the stability constants (less than 0.5 order of magnitude in most cases). The first complex is responsible for the pH changes observed at acid conditions ($\text{pH} \leq 3.5$), the second one dominates at pH ranging from 5 to 7, and both complexes coexist in the pH range 3.5–5. The stoichiometry and charge of these complexes are in agreement with those reported by Evdokimov and Kogan (1971) and Clark and Waddams (1957) and with those for the Ge complexes formed with analogous tartaric, lactic, and malic acids (Vartapetian, 1957). The stability constants for these complexes corresponding to the reactions



were calculated for each injection of the Ge-bearing titrant solution into the solution of citric acid. The values of $\text{Ge(OH)}_4(\text{H}_3\text{Cit})_2^{2-}$ and $\text{Ge(OH)}_4(\text{H}_2\text{Cit})^{2-}$ stability constants determined in this study at 25 and 90°C are listed in Table 6. These values are independent, in the limit of their uncertainties (± 0.3 log at 25°C and ± 0.1 log at 90°C), of pH and Ge concentration. The average values of the stability constants for both complexes, calculated at 25 and 90°C as the arithmetic

Table 7. Complexes between $\text{Ge}(\text{OH})_4$ and citric acid (H_4Cit), adopted in the treatment of the potentiometric measurements

Formula	Stoichiometry Ge: Cit	Charge
$\text{Ge}(\text{OH})_4(\text{H}_4\text{Cit})^0$	1:1	0
$\text{Ge}(\text{OH})_4(\text{H}_3\text{Cit})^-$	1:1	-1
$\text{Ge}(\text{OH})_4(\text{H}_2\text{Cit})^{2-}$	1:1	-2
$\text{Ge}(\text{OH})_4(\text{HCit})^{3-}$	1:1	-3
$\text{Ge}(\text{OH})_4(\text{Cit})^{4-}$	1:1	-4
$\text{Ge}(\text{OH})_4(\text{H}_4\text{Cit})_2^0$	1:2	0
$\text{Ge}(\text{OH})_4(\text{H}_4\text{Cit})(\text{H}_3\text{Cit})^-$	1:2	-1
$\text{Ge}(\text{OH})_4(\text{H}_3\text{Cit})_2^{2-}$	1:2	-2
$\text{Ge}(\text{OH})_4(\text{H}_3\text{Cit})(\text{H}_2\text{Cit})^{3-}$	1:2	-3
$\text{Ge}(\text{OH})_4(\text{H}_2\text{Cit})_2^{4-}$	1:2	-4
$\text{Ge}(\text{OH})_4(\text{H}_2\text{Cit})(\text{HCit})^{5-}$	1:2	-5
$\text{Ge}(\text{OH})_4(\text{HCit})_2^{6-}$	1:2	-6

mean of the values determined for each Ge and citric acid concentration, are also listed in Table 6. The standard molal enthalpy and entropy of reactions 3 and 4 were calculated assuming $\Delta_r\text{H}^\circ$ was constant: $\Delta_r\text{H}^\circ = -11.65$ kcal/mol and $\Delta_r\text{S}^\circ = -11.40$ cal/(mol.K); $\Delta_r\text{H}^\circ = -5.87$ kcal/mol and $\Delta_r\text{S}^\circ = -8.89$ cal/(mol.K).

3.2.2. Solubility measurements

The results of $\text{GeO}_2(\text{hex})$ solubility measurements performed at 25°C in NaCl-NaOH (or HCl) solutions and in the presence of citric, oxalic, tartaric, salicylic, acetic, and humic

acids are reported in Table 8. Samples were taken each 3–5 days. For most runs the steady-state for Ge concentration and pH was reached in 2 weeks and remained constant during at least a month. The values of Ge concentration at steady-state are independent of the solution/solid ratio for solutions with the same ligand concentration and pH. The attainment of dissolution steady-state was used as the equilibrium criterion. Solubilities measured in solutions without organic compounds at acid and neutral pH are in excellent agreement with numerous literature data (see Pokrovski and Schott (1998) and references therein). Unlike silica solubility, $\text{GeO}_2(\text{hex})$ solubility increases significantly in the presence of citric, oxalic, and tartaric acids. Furthermore, this increase is associated with strong pH changes between the initial and final solutions. These results clearly demonstrate the formation of Ge complexes with the above carboxylic acids. To determine the stoichiometry and stability of these complexes, we applied the rules already used in the potentiometric study of Ge-citric acid complexation (see above). The complexes likely to form were introduced into a BALANCE input file and their stability constants were systematically adjusted until calculated solubilities and pH equaled the measured values. The derived stability constants corresponding to the reactions given below in the text are listed in Table 8.

In the case of citric acid (H_4Cit), the complexes formed according to reactions 3 and 4 are responsible for the increase of solubility and the change of pH in the runs performed in acid medium ($\text{pH} < 4$). Their stability constants are in reasonable agreement with those derived from the potentiometric study

Table 8. $\text{GeO}_2(\text{hex})$ solubility at 25°C in NaCl-HCl/NaOH solutions in the presence of citric, tartaric, oxalic, acetic, salicylic and humic acids—Initial and final solution pH values and logarithms of the equilibrium constants of reactions (3)–(8)

Solution composition (mol/kg H_2O)	pH initial	pH final	m_{Ge} (mol/kg H_2O)	log K
Citric acid 0.102 m	3.72	2.99	0.087 ± 0.003	(3):4.65
NaOH 0.102 m				(4):2.97
Citric acid 0.036 m	3.79	3.26	0.058 ± 0.002	(3):6.11
NaOH 0.036 m				(4):2.92
Citric acid 0.103 m	6.28	6.90	0.16 ± 0.02	(4):2.25
NaOH 0.28 m				(5):2.30
Citric acid 0.020 m	6.33	6.70	0.060 ± 0.006	(4):1.80
NaOH 0.054 m + NaCl 0.08 m				(5):2.20
Citric acid 0.096 m	6.86	7.05	0.18 ± 0.01	(4):2.00
NaOH 0.279 m				(5):2.60
			log K (3–5) average	(3):5.4 $\pm$ 0.9
				(4):2.4 $\pm$ 0.6
				(5):2.4 $\pm$ 0.2
Tartaric acid 0.101 m	3.49	3.04	0.133 ± 0.002	(6):2.70
NaOH 0.100 m				(7):2.51
Oxalic acid 0.092 m	2.94	4.03	0.079 ± 0.003	(8):4.37
NaOH 0.10 m + NaCl 0.089 m				
Oxalic acid 0.101 m	4.84	5.60	0.045 ± 0.003	(8):4.33
NaOH 0.195 m				
Salicylic acid 0.10 m	6.45	6.30	0.039 ± 0.002	—
NaOH 0.101 m				
Acetic acid 0.099 m	4.62	4.61	0.042 ± 0.001	—
NaOH 0.047 m				
Humic acid 1 g COD/l	7.15	6.90	0.043 ± 0.003	—
NaOH + HCl				
NaOH 0.001 m	11.10	7.00	0.046 ± 0.002	—
NaCl 0.1 m	4.80	4.60	0.042 ± 0.003	—
NaCl 0.3 m	4.70	4.50	0.041 ± 0.002	—
NaCl 0.1 m + HCl 0.01 m	2.05	2.05	0.043 ± 0.002	—

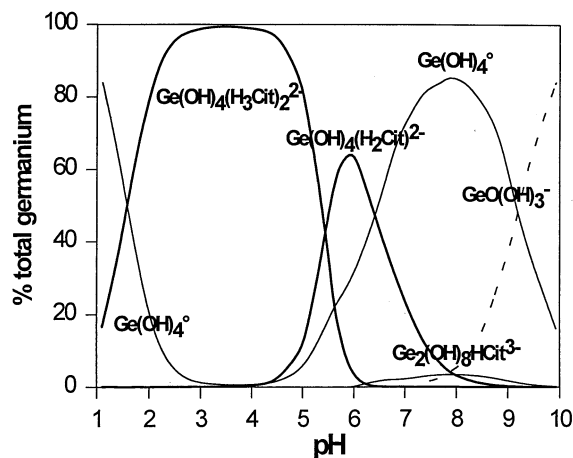


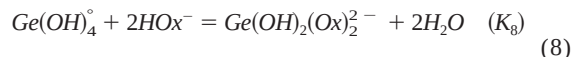
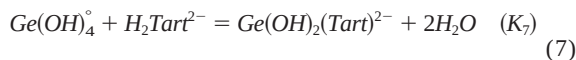
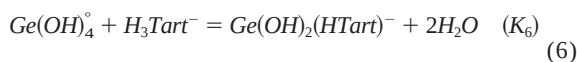
Fig. 3. Distribution of Ge aqueous species at 25°C in the H^+ - Ge(OH)_4 -citric acid system (0.005 m Ge(OH)_4 + 0.02 m citric acid + 0.1 m NaCl). It can be seen that Ge-citrate complexes (formed according to reactions 3 and 4) dominate Ge speciation in the pH range 2–7.

(Table 6), but are affected by a greater uncertainty because of the small number of solubility runs. At $\text{pH} > 6$, $\text{GeO}_2(\text{hex})$ solubility is too high to be explained solely by the formation of 1:1 complexes between Ge(OH)_4 and citrate. The high aqueous Ge concentrations suggest the presence of 2:1 Ge-citrate complexes formed according to



The equilibrium constants of reaction 5 derived for each experimental run and their mean value are listed in Table 8. Using this value, together with those for K_3 and K_4 derived from the present potentiometric study, we calculated the distribution of Ge-bearing species as a function of pH at 25°C in a solution of citric acid analogous to those used in our potentiometric measurements (0.005 m Ge(OH)_4 + 0.02 m H_4Cit + 0.1 m NaCl; Fig. 3). It can be seen that the complex $\text{Ge}_2(\text{OH})_8\text{HCit}^{3-}$, which is responsible for the high $\text{GeO}_2(\text{hex})$ solubilities measured at neutral pH, accounts only for 2–3% of total Ge in the solutions used for potentiometric measurements. This explains why it was not detected in this potentiometric study. In the pH range 2–7, complexes formed with the anions of citric acid H_3Cit^- and $\text{H}_2\text{Cit}^{2-}$ dominate Ge speciation. This is consistent with the solubility study of Evdokimov and Kogan (1971) which implies the formation of a 1:2 Ge-citrate complex similar to that formed by reaction 3. At $\text{pH} > 7$ Ge-citrate complexes become less stable. This is in agreement with the measurements of Vartapetian (1957) who did not observe, at $\text{pH} > 7$, Ge(OH)_4° complexation with citric acid analogs like tartaric, lactic, and malic acids.

In the case of tartaric (H_4Tart) and oxalic (H_2Ox) acids, the following reactions have been found to account for the solubility increase and pH change observed in the $\text{GeO}_2(\text{hex})$ solubility experiments:



The values of the stability constants of these complexes are reported in Table 8. Reaction 6 is the same as that proposed by Vartapetian (1957) who found $\log Q_6 = 5.2$ in a 1 M NaCl solution at 18°C. The stoichiometry of the Ge-oxalic acid complex derived in this study, $\text{Ge(OH)}_2(\text{C}_2\text{O}_4)_2^{2-}$, is the same as that proposed by Everest (1955) in the pH range 3–6. Note that the consideration of other complexes such as $\text{Ge(OH)}_4(\text{C}_2\text{O}_4)^{2-}$ and/or $\text{Ge}(\text{C}_2\text{O}_4)_3^{2-}$ (Moeller and Nielsen, 1953; Everest, 1955; Tananaev and Avduevskaya, 1958) does not enable to reproduce the solubility increase observed in this study.

In the presence of acetic (0.1 m acid + 0.05 m NaOH) and salicylic (0.1 m acid + 0.1 m NaOH) acids, $\text{GeO}_2(\text{hex})$ solubilities are the same as those measured in the 0.1 m NaCl solutions, suggesting that there is no significant complexation of aqueous Ge. Similarly, there was no solubility increase in the presence of humic acid (1g/L DOC). It should be noted, however, that the concentration of carboxylic and phenolic groups ($\sim 10^{-3}$ mol/g of acid) in this experiment was probably too low for Ge-humate complexation to distinctly affect the high aqueous Ge concentrations (0.04 m) in equilibrium with $\text{GeO}_2(\text{hex})$.

4. COMPARISON OF THE COMPLEXATION OF GE AND SI WITH AQUEOUS ORGANIC SPECIES

4.1. Nature and Stability of the Complexes

4.1.1. Germanium

The results obtained in this study, together with available literature data, show that Ge is able to form stable complexes with organic ligands having hydroxyl-alcoholic, hydroxyl-phenolic, and carboxylic functional groups. The key for the formation of such complexes is the establishment of bidentate covalent bonds (of chelate type) leading to the formation of pentagonal or hexagonal rings which incorporate two Ge-O-C bonds. This requires that Ge enlarges its coordination number from 4 to 6 and that the organic ligand has at least two functional groups attached to two different carbons. The negative values of the enthalpy and entropy of formation of these complexes are consistent with a chelate structure, reflecting its compactness and the fact that the strength of covalent donor-acceptor bonds decreases with increasing temperature (Tandura et al., 1986).

In the case of Ge complexation with glycols and polyalcohols, the bonds are formed with two hydroxyl groups of the alcohol molecule located in the *cis*-position. Complexes with benzene derivatives (quinoleines, pyrons, benzoquinones) are formed with one hydroxyl and one carbonyl ($\text{C}=\text{O}$) group. The stoichiometry of these complexes is 1:1 or/and 1:2, depending on steric constraints (group positions, inter atomic distances, and angles) (Choux and Benoit, 1967; Antikainen, 1959). With catechol, which exhibits a higher complexing capacity (Sillen and Martell, 1971), Ge forms a 1:3 complex whose structure is given in Fig. 4a. Note that boron and arsenic form only 1:1 and

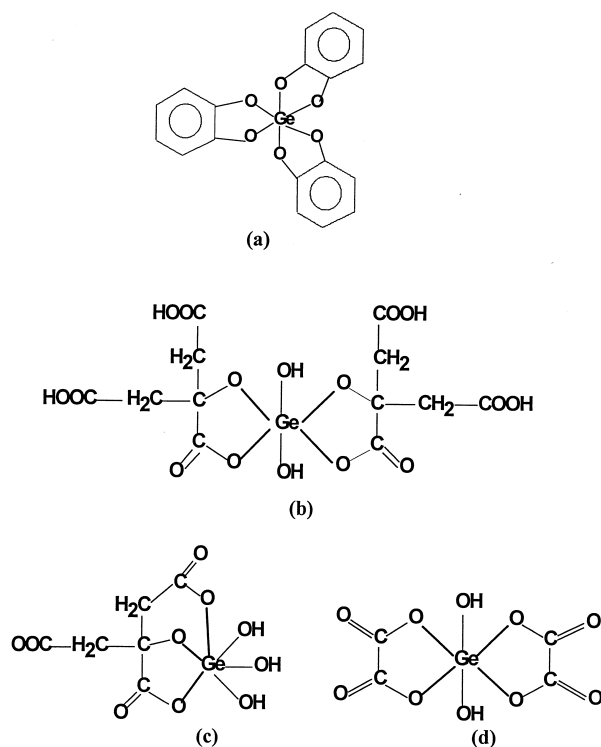
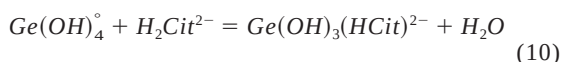
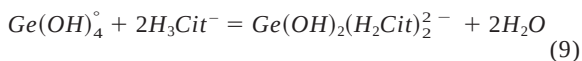


Fig. 4. Stylized figures of Ge-catechol (a), Ge-citric acid (b and c) and Ge-oxalic acid (d) complexes.

1:2 complexes with catechol because their coordination number cannot be higher than 4 (Cotton and Wilkinson, 1966).

The existence of Ge complexes with di- and tri-carboxylic acids, and hydroxy-carboxylic acids (oxalic, tartaric, citric, lactic, etc.) together with the absence of complexation with monocarboxylic acids (acetic) is also consistent with the formation of bi- or tridentate chelates constituted of five- or six-membered rings with two functional groups (carboxyls and/or hydroxyls) in the *cis*- or *meta*-position, respectively (Martell and Hancock, 1996). The facility of Ge to enlarge its coordination sphere to 6 can explain the fact that its complexes with carboxylic acids are more stable than those formed by boron and arsenic (Sillen and Martell, 1971). The required sixfold coordination of Ge for these complexes formation has been recently confirmed by our EXAFS and XANES study of aqueous Ge-bearing solutions in the presence of oxalic and citric acids and catechol (Pokrovski et al., 1999). XANES and EXAFS spectra of these solutions are consistent with 6 oxygen atoms around a Ge atom in its first coordination shell, with Ge-O distances from 1.85 to 1.93 Å. Consequently, reactions 3 and 4 have been rewritten to account for the chelate structure of hexacoordinated Ge-citrate complexes:

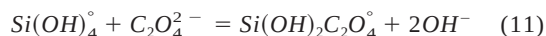


The schematic structures of the Ge complexes formed with citric acid (reactions 9 and 10) and with oxalic acid (reaction 8)

are given in Fig. 4b,c, and d, respectively. These structures are analogous to those formed by other metals and metalloids with citrate and oxalate (Martell and Hancock, 1996).

4.1.2. Silicon

The results of this study show that Si, unlike Ge, exhibits a very weak affinity to carboxylic acids and alcohols in aqueous solution. Based on their observation of an increase of quartz dissolution rate and solubility in the presence of citrate and oxalate at neutral pH, Bennett et al. (1988) and Bennett (1991) proposed that silica forms both aqueous and surface complexes with these acids. They postulated the formation of esters according to the reaction:



This reaction implies that the formation of Si-oxalate or Si-citrate complexes should yield significant changes in solution pH. However, the potentiometric measurements reported in this study and in those of Öhman et al. (1991) and Poulson et al. (1997) have not detected any pH shifts in the presence of up to 0.4 M of carboxylic acid. These data are not consistent with the formation of strong ester-bonded complexes as suggested by Bennett. In view of their weak stability, it is more likely that silica-carboxylic or alcoholic complexes are formed via hydrogen bonding (Öhman et al., 1991). In contrast, the results reported in this study and in those of Öhman et al. (1991), Sedeh et al. (1992, 1993) and Sjöberg et al. (1985) show that silica can form chelate-type complexes with pyrocatechol (reaction 1) and its derivatives, substituted ortho-diphenols, and tropolones. In these complexes, as it was demonstrated by NMR and X-ray measurements (Flynn and Boer, 1969; Sjöberg et al., 1985; Sedeh, 1992), Si is hexa-coordinated and forms five membered bidentate rings like Ge in Ge-catechol complex (Fig. 4a). The stability constant of Si-catechol complexes at 25°C ($\log K_1 = -12.0$), however, is 10 orders of magnitude lower than that of their Ge analogs ($\log K_2 = -1.4$). Similar differences in stability are also observed between analogous Si and Ge complexes formed with other substituted ortho-diphenols (for example, the values of $\log K$ for $\text{Si}(\text{Pyrogallol})_3^{2-}$ and $\text{Ge}(\text{Pyrogallol})_3^{2-}$ complexes are -10.02 (Sedeh et al., 1992) and -1.33 (Andrianov and Nazarenko, 1963), respectively). The weak affinity of silica for aqueous organic ligands results both from steric strains affecting the complexes in which Si is tetracoordinated and from the reluctance of Si, unlike Ge, to enlarge its coordination sphere to form hexacoordinated complexes.

As is the case for metals, the critical step in the formation of Si-organic complexes is the establishment of a five-membered chelate ring between Si and two adjacent functional groups of the organic ligand. For such a ring to be stable requires the metal coordination number to be equal to 6, as it is for Al or Fe, but not 4 like for silica. Indeed, the tetrahedral fourfold coordination induces an angular distortion in the formed chelate which gives rise to strain energy (Bennett, 1991; Martell and Hancock, 1996). Molecular orbital calculations for bidentate Si-organic complexes indicate that the five-membered ring structure is the most stable configuration when Si is hexa- or pentacoordinated with Si-O-Si bond angles of 90° (Bennett and Casey, 1994). This result is confirmed by the X-ray diffraction

Table 9. Stability constants at 25°C of aqueous Ge and Si complexes with catechol, citric acid and fructose

Reaction	log K	Reference
Ge		
<i>Catechol (H₂L)</i>		
$\text{Ge}(\text{OH})_4^0 + 3\text{H}_2\text{L}^0 = \text{GeL}_3^{2-} + 2\text{H}^+ + 4\text{H}_2\text{O}$	-1.39	This study
<i>Citric acid (H₄L)</i>		
$\text{Ge}(\text{OH})_4^0 + 2\text{H}_3\text{L}^- = \text{Ge}(\text{OH})_2(\text{H}_2\text{L})_2^{2-} + 2\text{H}_2\text{O}$	6.2	This study
$\text{Ge}(\text{OH})_4^0 + \text{H}_2\text{L}^{2-} = \text{Ge}(\text{OH})_3\text{HL}^{2-} + \text{H}_2\text{O}$	2.4	This study
<i>Fructose (L)</i>		
$\text{GeO}(\text{OH})_3^- + 2\text{L}^0 = \text{Ge}(\text{OH})(\text{H}_2\text{L})_2^- + 3\text{H}_2\text{O}$	5.48	Sillen and Martell (1971)
Si		
<i>Catechol (H₂L)</i>		
$\text{Si}(\text{OH})_4^0 + 2\text{H}_2\text{L}^0 = \text{SiL}_2^{2-} + 2\text{H}^+ + 4\text{H}_2\text{O}$	-12.0	This study

studies of Si-catechol complexes, which show that Si is located at the centre of a slightly deformed bi-pyramid with six oxygen atoms at its apices. The O-Si-O angles and Si-O distances in this bi-pyramid which range from 86 to 94°, and from 1.77 to 1.81 Å, respectively (Flynn and Boer, 1969), differ significantly from those in the Si(OH)₄ tetrahedron which are equal to 104° and 1.65 Å, respectively (Kubicki et al., 1993). However, it is well known that, unlike Ge which has a bigger ionic radius (0.39 Å for Ge⁴⁺ vs. 0.26 Å for Si⁴⁺), Si exhibits a coordination four in its compounds with oxygen (oxides, silicates), in conformity with Pauling's rules. Octahedral coordination is observed only in high-pressure phases (stishovite, MgSiO₃-ilmenite). Thus, the tendency for Si not to be octahedrally coordinated by oxygen atoms explains the weak stability of its complexes with aqueous organic species.

4.1.3. Comparison of the stability of aqueous Ge- and Si-organic complexes

To compare the relative stability of Ge and Si complexes formed with the main functional groups of organic compounds, we have calculated, as a function of pH, the distribution of Ge and Si-organic species in three solutions containing 10⁻⁴ m of total Ge or Si and either 10⁻³ m of catechol, citric acid, or polyalcohol (fructose). Fructose was selected to illustrate the maximal capacity of Ge complexing by alcoholic hydroxide groups (Antikainen, 1959). The stability constants used in these calculations were taken from this study for the Ge-catechol, Si-catechol and Ge-citric acid complexes and from Sillen and Martell (1971) for the Ge-fructose complex. They are listed in Table 9 together with the corresponding complex formation reactions. The distribution of Ge-organic species deduced from this calculation is shown in Fig. 5. In this figure, the distribution curves of the Ge-organic complexes in each solution are superposed to compare the complexing capacity of the different functional groups. It can be seen that the complexes formed with ortho-phenolic compounds are the most stable and dominate Ge speciation at pH > 5. Complexes with carboxylic acids are less stable and form only at acid conditions (pH < 6) whereas those formed with polyalcohols develop only in basic solutions (pH > 9). Silicon is not complexed by carboxylic acids nor polyalcohols, and the complexes formed with catechol account for less than 0.1% of total aqueous silica.

4.2. Role of Natural Organic Acids on Si and Ge Mobility, and Ge/Si Ratios in Superficial Waters

The data generated in this study strongly suggest that the complexation of Ge with aqueous humic and fulvic acids, which have the same functional groups as the simple organic ligands, could be important in natural waters. The extent of Ge complexation with a typical humic acid found in superficial waters (5 meq/g of carboxyl, 2 meq/g aromatic hydroxyl, and 7 meq/g aliphatic hydroxyl groups, see Chapter 10 in Thurman, 1985) can be roughly estimated by assuming that the stability constants of Ge complexes formed with the humic acid different functional groups are the same as those with the corresponding simple organic compounds having the same groups. Such calculations show that in a solution containing 20 mg/L of DOC and 10⁻⁹ m of Ge, typical encountered concentrations in tropical watersheds and rivers (Thurman, 1985; Viers et al., 1997), complexes with aromatic hydroxyl groups (catechol type) can account for more than 95% of the total dissolved Ge at pH ≥ 6–7. The carboxylic groups of humic acids can also

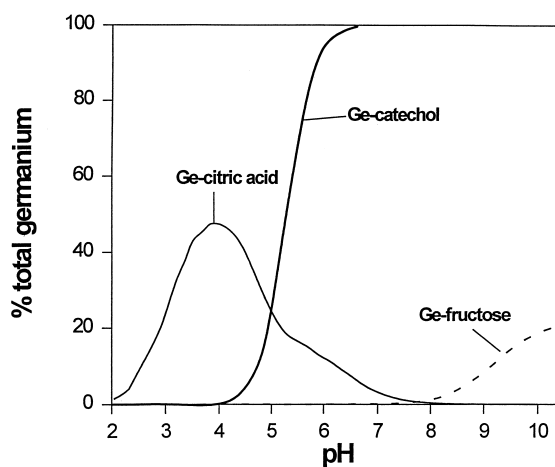


Fig. 5. Distribution curves of Ge-organic complexes as a function of pH at 25°C in aqueous solutions containing 0.001 m catechol, or 0.001 m citric acid or 0.001 m glucose. The curves are superposed to compare the complexing capacity of these organic ligands. Germanium forms complexes of chelate type with the following functional groups: (1) carboxylic in acid media (2 ≤ pH ≤ 6), (2) di-phenolic hydroxyls at neutral to basic pH (pH ≥ 5–6), and (3) alcoholic hydroxyls in basic solutions (pH ≥ 9).

Table 10. Concentrations of Si, Ge, and dissolved organic carbon (DOC) measured in the waters of the Nsimi-Zoétélé watershed (Cameroon) after successive filtrations through membranes of decreasing pore size from 0.22 to 0.003 μm (according to Viers et al., 1997)

	0.22 μm	0.025 μm	0.003 μm
pH = 4.6; T = 21.8°C			
Si (mg/L)	8.55	8.57	8.27
Ge ($\mu\text{g/L}$)	0.078 ± 0.037	<0.005	<0.005
DOC (mg/L)	23.76	7.71	3.22
pH = 5.5; T = 22.6°C			
Si (mg/L)	18.60	18.32	17.72
Ge ($\mu\text{g/L}$)	0.047 ± 0.023	<0.005	<0.005
DOC (mg/L)	18.07	14.46	2.39

Uncertainties are given for the two sigma standard deviation.

contribute to Ge complexation ($\sim 10\%$) in slightly acidic waters ($4 \leq \text{pH} \leq 6$). In contrast, complexes with alcoholic hydroxyls are negligible. It should be noted that the distribution of Ge complexes between carboxylic and aromatic sites can vary considerably depending on the nature of the aquatic humic substances. For example, in the presence of fulvic acids which generally contain a higher fraction of carboxylic groups and less aromatic sites, Ge complexation would be stronger in acidic (up to 20% of total Ge) and weaker in neutral waters. Thus, in rivers with low DOC amount ($< 2\text{--}5 \text{ mg/L}$) which is constituted essentially of fulvic acids, Ge complexation is likely to be weak. On the other hand, in bog, swamp and lake waters where dissolved humic substances are enriched in phenolic hydroxyl groups (Thurman, 1985) complexes of catechol type dominate Ge speciation from slightly acidic to basic pH.

Similar calculations carried out for Si in the same solution containing 20 mg/L of DOC and 10^{-4} m of Si, and assuming that the stability constant of the complexes formed by Si with the phenolic OH-sites of the humic acid is equal to that of Si-catechol complexes ($\log K_1 = -12.0$), suggest that Si-humic acid complexation is negligible ($< 0.1\%$ of total Si). It should be noted that the stability of the complexes formed by Si with catechol derivatives like protocatechuic and gallic acids are 1–1.5 order of magnitude higher than that of Si-catechol complexes (Sedeh et al., 1992). Nevertheless, even if the stability constant for Si-phenolic OH sites were increased by 2 orders of magnitude, Si-humic acid complexes would account for less than 1% of the total Si present in solution.

Our predictions of Ge and Si complexation with humic substances are supported by the results of ultrafiltrations carried out by Viers et al. (1997) on the organic-rich ($10 \text{ mg/L} \leq \text{DOC} \leq 30 \text{ mg/L}$) waters of the Zsimi-Zoétélé watershed in Cameroon. In Table 10 are listed the concentrations of DOC, Si and Ge measured in these waters after their successive filtrations through membranes of decreasing pore size from 0.2 to 0.003 μm . It can be seen in this Table that DOC concentrations decrease significantly after each filtration which demonstrates the retention of high molecular mass humic substances on the filter. Silica concentration is not changed by successive filtrations showing the absence of Si-humic acid complexation. By contrast, after the first filtration, Ge concentration falls below its detection limit by ICP-MS which can be explained by the retention of the large Ge-humate complexes on the membrane filters. The presence of dissolved organic matter can thus con-

siderably affect Ge concentration and increase Ge/Si ratios in organic-rich natural waters. For example, the average Ge/Si molal ratio in the Zsimi-Zoétélé waters (10 mg/L of DOC) is about $4 \cdot 10^{-6}$. This value is one order of magnitude higher than the global mean Ge/Si ratio in rivers ($0.6 \cdot 10^{-6}$, Mortlock and Froelich, 1987) which have a much lower DOC content ranging from 0.5 to 2 mg/L (Thurman, 1985).

The results of the present study indicate that dissolved organic matter can markedly affect the superficial cycle of Ge but not that of Si, although these two elements are generally considered as chemical analogs following the same geochemical pathways (Froelich and Andreae, 1981; Froelich et al., 1985a,b; Froelich et al., 1992). Continental weathering and mid-ocean-ridge hydrothermal activity are the two main processes that release dissolved Si and Ge to the ocean. It has been hypothesized that temporal variations in the relative intensity of these processes might be estimated from the Ge/Si ratio of opals in marine sediments because inorganic Ge is incorporated into biogenic opal as if it were a heavy isotope of Si (Froelich and Andreae, 1981). More recently, it has been suggested from analyses of river water in the Orinoco basin that chemical weathering intensity on the continent can be related to rivers Ge/Si ratios (Murnane and Stallard, 1990). The results obtained in this study imply that the Ge/Si ratios measured in organic-rich rivers should be used with caution to estimate chemical-weathering intensity because they can be significantly increased due to Ge complexation with organic matter. Moreover, there is ample evidence that the biological uptake of different trace elements from solution is controlled essentially by their aqueous speciation. Many metals and metalloids present as aqueous complexes with inorganic and organic ligands are generally less available for organisms than when in the form of free cations or hydroxide complexes (Andreae, 1979; Apte et al., 1986; Lewis et al., 1988; Birshall et al., 1989; and references therein). It can be thus hypothesized that the uptake by diatoms of aqueous Ge complexed with organic matter will be significantly lower than that of noncomplexed germanic acid. As a result, the Ge/Si ratios determined in marine opal sediments formed in organic-rich waters are not likely to reflect the Ge/Si ratio in seawater environment. Knowledge of the amount and nature of dissolved organic matter should help to interpret measured Ge/Si ratios in different environments and to unambiguously use Ge as a tracer of surficial geochemical processes.

5. CONCLUSION

The stability of the complexes formed by Si and Ge with aqueous organic species having functional groups typical of natural dissolved organic matter (carboxyl, alcoholic hydroxyl, and phenolic hydroxyl) was determined from 25 to 90°C via solubility and potentiometric methods. The results obtained in this study show that Ge forms stable hexacoordinated complexes of chelate type with di- and polyfunctional carboxylic acids, polyalcohols and ortho-diphenols whereas analogous complexes formed by silica are very weak. The absence of stable Si complexes with aqueous organic ligands can be explained both by steric strains affecting the complexes in which Si is tetracoordinated, and by the reluctance of Si, unlike Ge, to enlarge its coordination sphere to form hexacoordinated Si-O structures.

The results obtained in this study have been applied to model Ge and Si complexing with humic acids assuming that the stability constants of Si and Ge complexes formed with the carboxylic, phenolic-hydroxyl, and alcoholic-hydroxyl functional sites of humic acids are the same as those with the corresponding simple organic compounds having the same functional groups. Calculations show that Si-humic acid complexes are negligible in most natural waters. In contrast, the presence of humic acids can considerably affect aqueous Ge speciation. The formation of Ge-humic acid complexes leads to a notable increase of Ge/Si ratios in organic-rich surficial waters and explains the parallel drops of Ge and DOC concentrations during successive ultrafiltrations of organic-rich waters (Viers et al., 1997). Germanium complexation with organic matter should be, thus, taken into account when using the Ge/Si ratios measured in surface waters and biogenic opals to estimate chemical-weathering intensity and Ge and Si global fluxes.

The lack of complexation between aqueous Si and organic ligands implies that no significant adsorption of organic acids can occur onto quartz surfaces and facilitate the disruption of Si-O-Si framework bonds. As a result, organic acids, which can notably increase the rate of dissolution of metal oxides and aluminosilicates via the formation of metal-organic surface and aqueous complexes (Furrer and Stumm, 1986; Oelkers and Schott, 1998), should have little effect upon quartz weathering rates.

Acknowledgements—This research has been supported by the French Ministère de l'Enseignement Supérieur et de la Recherche (awarding of a Ph. D. fellowship to G. P.) and by a grant from CNRS/PIRSEM (Arc Métallurgie). We are grateful to J. Escalier, B. Reynier, J.-M. Gautier, and M. Thibaut for their help in chemical analyses, and J.-C. Harri-choury and C. Lurde for their technical assistance during the experiments. We would like to thank J.-L. Dandurand, J. Viers, B. Dupré, F. Martin, and S. Deberdt for discussions on the captivating problems of metal-organic complexation. This paper benefited greatly from the reviews of J. Fein and two anonymous reviewers.

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